



MATH110: Hypothesis Testing

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Learning Objectives

- State null and alternative hypotheses
- Compute a one-sample z test statistic
- Make a decision at a given significance level

Simplify each expression completely. Show all steps and circle your final answer.

Chi-Square Goodness of Fit

1. Observed frequencies: [18, 22, 20, 15, 25]. Expected frequencies: [20, 20, 20, 20, 20]. Compute the chi-square goodness-of-fit test statistic.

$$O = [18, 22, 20, 15, 25], \quad E = [20, 20, 20, 20, 20]$$

Answer: _____

2. Observed frequencies: [30, 25, 20, 15, 10]. Expected frequencies: [20, 20, 20, 20, 20]. Compute the chi-square goodness-of-fit test statistic.

$$O = [30, 25, 20, 15, 10], \quad E = [20, 20, 20, 20, 20]$$

Answer: _____

3. Observed frequencies: [12, 18, 24, 22, 24]. Expected frequencies: [20, 20, 20, 20, 20]. Compute the chi-square goodness-of-fit test statistic.

$$O = [12, 18, 24, 22, 24], \quad E = [20, 20, 20, 20, 20]$$

Answer: _____

4. Observed frequencies: [18, 22, 20, 15, 25]. Expected frequencies: [20, 20, 20, 20, 20]. Compute the chi-square goodness-of-fit test statistic.

$$O = [18, 22, 20, 15, 25], \quad E = [20, 20, 20, 20, 20]$$

Answer: _____

Chi-Square Test for Independence

5. A 2x2 contingency table has counts: 12, 28 (row 1) and 19, 39 (row 2). Test for independence. Compute the chi-square statistic.

Row1: 12, 28 Row2: 19, 39

Answer: _____



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6. A 2x2 contingency table has counts: 27, 13 (row 1) and 42, 40 (row 2). Test for independence. Compute the chi-square statistic.

Row1: 27, 13 Row2: 42, 40

Answer: _____

7. A 2x2 contingency table has counts: 41, 21 (row 1) and 45, 23 (row 2). Test for independence. Compute the chi-square statistic.

Row1: 41, 21 Row2: 45, 23

Answer: _____

8. A 2x2 contingency table has counts: 17, 28 (row 1) and 39, 38 (row 2). Test for independence. Compute the chi-square statistic.

Row1: 17, 28 Row2: 39, 38

Answer: _____

Matched-Pairs t-Test

9. A matched-pairs study yields differences: [3, 5, 2, 4, 1, 3, 2, 4]. Test $H_0: \mu_d = 0$ at $\alpha = 0.05$. Compute the t-statistic.

$d = [3, 5, 2, 4, 1, 3, 2, 4]$, $\bar{d} = 3.0$

Answer: _____

10. A matched-pairs study yields differences: [5, 3, -1, 4, 2, 6, 1, 4]. Test $H_0: \mu_d = 0$ at $\alpha = 0.05$. Compute the t-statistic.

$d = [5, 3, -1, 4, 2, 6, 1, 4]$, $\bar{d} = 3.0$

Answer: _____

11. A matched-pairs study yields differences: [5, 3, -1, 4, 2, 6, 1, 4]. Test $H_0: \mu_d = 0$ at $\alpha = 0.05$. Compute the t-statistic.

$d = [5, 3, -1, 4, 2, 6, 1, 4]$, $\bar{d} = 3.0$

Answer: _____

12. A matched-pairs study yields differences: [-2, 1, -3, 0, 2, -1, 1, -2]. Test $H_0: \mu_d = 0$ at $\alpha = 0.05$. Compute the t-statistic.

$d = [-2, 1, -3, 0, 2, -1, 1, -2]$, $\bar{d} = -0.5$

Answer: _____

Type I and Type II Errors



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13. Identify the type of error: H_0 : A new drug has no effect. A researcher rejects H_0 when in fact the drug truly has no effect.

Answer: _____

14. Identify the type of error: H_0 : A machine is working correctly. An inspector fails to reject H_0 even though the machine is actually broken.

Answer: _____

15. Identify the type of error: H_0 : A student is not cheating. A professor rejects H_0 and accuses the student, but the student was actually innocent.

Answer: _____

16. Identify the type of error: H_0 : A new drug has no effect. A researcher rejects H_0 when in fact the drug truly has no effect.

Answer: _____

17. Identify the type of error: H_0 : A machine is working correctly. An inspector fails to reject H_0 even though the machine is actually broken.

Answer: _____

18. Identify the type of error: H_0 : A student is not cheating. A professor rejects H_0 and accuses the student, but the student was actually innocent.

Answer: _____

19. Identify the type of error: H_0 : A new drug has no effect. A researcher rejects H_0 when in fact the drug truly has no effect.

Answer: _____

20. Identify the type of error: H_0 : A machine is working correctly. An inspector fails to reject H_0 even though the machine is actually broken.

Answer: _____

21. Identify the type of error: H_0 : A student is not cheating. A professor rejects H_0 and accuses the student, but the student was actually innocent.

Answer: _____

Two-Sample z-Test for Means

22. Group 1: $\bar{x}_1=95$, $\sigma_1=9$, $n_1=100$. Group 2: $\bar{x}_2=80$, $\sigma_2=14$, $n_2=36$. Test H_0 : $\mu_1=\mu_2$ at $\alpha=0.05$.

$$\bar{x}_1 = 95, \bar{x}_2 = 80, \sigma_1 = 9, \sigma_2 = 14, n_1 = 100, n_2 = 36$$

Answer: _____



23. Group 1: $\bar{x}_1=62$, $\sigma_1=5$, $n_1=100$. Group 2: $\bar{x}_2=83$, $\sigma_2=12$, $n_2=49$. Test H_0 : $\mu_1=\mu_2$ at $\alpha=0.05$.

$$\bar{x}_1 = 62, \bar{x}_2 = 83, \sigma_1 = 5, \sigma_2 = 12, n_1 = 100, n_2 = 49$$

Answer: _____

24. Group 1: $\bar{x}_1=64$, $\sigma_1=18$, $n_1=64$. Group 2: $\bar{x}_2=79$, $\sigma_2=12$, $n_2=49$. Test H_0 : $\mu_1=\mu_2$ at $\alpha=0.05$.

$$\bar{x}_1 = 64, \bar{x}_2 = 79, \sigma_1 = 18, \sigma_2 = 12, n_1 = 64, n_2 = 49$$

Answer: _____

One-sample z-test for a mean

25. Test H_0 : $\mu = 48$ against H_1 : $\mu \neq 48$. A sample of $n = 16$ gives a sample mean of 44 with $\sigma = 15$. Compute the z test statistic and state the decision at $\alpha = 0.05$.

$$H_0 : \mu = 48, \quad \bar{x} = 44, \quad \sigma = 15, \quad n = 16$$

Answer: _____

26. Test H_0 : $\mu = 114$ against H_1 : $\mu \neq 114$. A sample of $n = 25$ gives a sample mean of 105 with $\sigma = 10$. Compute the z test statistic and state the decision at $\alpha = 0.05$.

$$H_0 : \mu = 114, \quad \bar{x} = 105, \quad \sigma = 10, \quad n = 25$$

Answer: _____

27. Test H_0 : $\mu = 49$ against H_1 : $\mu \neq 49$. A sample of $n = 100$ gives a sample mean of 37 with $\sigma = 15$. Compute the z test statistic and state the decision at $\alpha = 0.05$.

$$H_0 : \mu = 49, \quad \bar{x} = 37, \quad \sigma = 15, \quad n = 100$$

Answer: _____

One-Sample z-Test for a Proportion

28. A researcher claims the true proportion is $p_0 = 0.5$. A sample of $n = 236$ gave 157 successes. Test H_0 : $p = 0.5$ at $\alpha = 0.05$.

$$H_0 : p = 0.5, \quad \hat{p} = 0.6653, \quad n = 236$$

Answer: _____

29. A researcher claims the true proportion is $p_0 = 0.6$. A sample of $n = 143$ gave 87 successes. Test H_0 : $p = 0.6$ at $\alpha = 0.05$.

$$H_0 : p = 0.6, \quad \hat{p} = 0.6084, \quad n = 143$$

Answer: _____



30. A researcher claims the true proportion is $p_0 = 0.7$. A sample of $n = 453$ gave 149 successes. Test $H_0: p = 0.7$ at $\alpha = 0.05$.

$$H_0 : p = 0.7, \quad \hat{p} = 0.3289, \quad n = 453$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Matched-Pairs t-Test, One-sample z-test for a mean, One-Sample z-Test for a Proportion, Two-Sample z-Test for Means, Chi-Square Test for Independence, Type I and Type II Errors, Chi-Square Goodness of Fit. All answers verified by independent computation.

Solutions



Chi-Square Goodness of Fit

1. Observed frequencies: [18, 22, 20, 15, 25]. Expected frequencies: [20, 20, 20, 20, 20]. Compute the chi-square goodness-of-fit test statistic.

$$O = [18, 22, 20, 15, 25], \quad E = [20, 20, 20, 20, 20]$$

→ Compute each $(O-E)^2/E$ term: $(18-20)^2/20=0.2$; $(22-20)^2/20=0.2$; $(20-20)^2/20=0.0$; $(15-20)^2/20=1.25$; $(25-20)^2/20=1.25$.

→ Sum all terms: chi-square = 2.9.

→ Degrees of freedom: $df = k - 1 = 4$.

Answer: $\chi^2 = 2.9, \quad df = 4$

2. Observed frequencies: [30, 25, 20, 15, 10]. Expected frequencies: [20, 20, 20, 20, 20]. Compute the chi-square goodness-of-fit test statistic.

$$O = [30, 25, 20, 15, 10], \quad E = [20, 20, 20, 20, 20]$$

→ Compute each $(O-E)^2/E$ term: $(30-20)^2/20=5.0$; $(25-20)^2/20=1.25$; $(20-20)^2/20=0.0$; $(15-20)^2/20=1.25$; $(10-20)^2/20=5.0$.

→ Sum all terms: chi-square = 12.5.

→ Degrees of freedom: $df = k - 1 = 4$.

Answer: $\chi^2 = 12.5, \quad df = 4$

3. Observed frequencies: [12, 18, 24, 22, 24]. Expected frequencies: [20, 20, 20, 20, 20]. Compute the chi-square goodness-of-fit test statistic.

$$O = [12, 18, 24, 22, 24], \quad E = [20, 20, 20, 20, 20]$$

→ Compute each $(O-E)^2/E$ term: $(12-20)^2/20=3.2$; $(18-20)^2/20=0.2$; $(24-20)^2/20=0.8$; $(22-20)^2/20=0.2$; $(24-20)^2/20=0.8$.

→ Sum all terms: chi-square = 5.2.

→ Degrees of freedom: $df = k - 1 = 4$.

Answer: $\chi^2 = 5.2, \quad df = 4$

4. Observed frequencies: [18, 22, 20, 15, 25]. Expected frequencies: [20, 20, 20, 20, 20]. Compute the chi-square goodness-of-fit test statistic.

$$O = [18, 22, 20, 15, 25], \quad E = [20, 20, 20, 20, 20]$$

→ Compute each $(O-E)^2/E$ term: $(18-20)^2/20=0.2$; $(22-20)^2/20=0.2$; $(20-20)^2/20=0.0$; $(15-20)^2/20=1.25$; $(25-20)^2/20=1.25$.

→ Sum all terms: chi-square = 2.9.

→ Degrees of freedom: $df = k - 1 = 4$.

Answer: $\chi^2 = 2.9, \quad df = 4$



Chi-Square Test for Independence

5. A 2x2 contingency table has counts: 12, 28 (row 1) and 19, 39 (row 2). Test for independence. Compute the chi-square statistic.

Row1: 12, 28 Row2: 19, 39

→ Row totals: row1=40, row2=58. Column totals: col1=31, col2=67. Grand total: N=98.

→ Expected counts: $E_{11}=12.65$, $E_{12}=27.35$, $E_{21}=18.35$, $E_{22}=39.65$.

→ $\chi^2 = (12-12.65)^2/12.65 + (28-27.35)^2/27.35 + (19-18.35)^2/18.35 + (39-39.65)^2/39.65 = 0.083$.

Answer: $\chi^2 = 0.083$

6. A 2x2 contingency table has counts: 27, 13 (row 1) and 42, 40 (row 2). Test for independence. Compute the chi-square statistic.

Row1: 27, 13 Row2: 42, 40

→ Row totals: row1=40, row2=82. Column totals: col1=69, col2=53. Grand total: N=122.

→ Expected counts: $E_{11}=22.62$, $E_{12}=17.38$, $E_{21}=46.38$, $E_{22}=35.62$.

→ $\chi^2 = (27-22.62)^2/22.62 + (13-17.38)^2/17.38 + (42-46.38)^2/46.38 + (40-35.62)^2/35.62 = 2.904$.

Answer: $\chi^2 = 2.904$

7. A 2x2 contingency table has counts: 41, 21 (row 1) and 45, 23 (row 2). Test for independence. Compute the chi-square statistic.

Row1: 41, 21 Row2: 45, 23

→ Row totals: row1=62, row2=68. Column totals: col1=86, col2=44. Grand total: N=130.

→ Expected counts: $E_{11}=41.02$, $E_{12}=20.98$, $E_{21}=44.98$, $E_{22}=23.02$.

→ $\chi^2 = (41-41.02)^2/41.02 + (21-20.98)^2/20.98 + (45-44.98)^2/44.98 + (23-23.02)^2/23.02 = 0.0$.

Answer: $\chi^2 = 0.0$

8. A 2x2 contingency table has counts: 17, 28 (row 1) and 39, 38 (row 2). Test for independence. Compute the chi-square statistic.

Row1: 17, 28 Row2: 39, 38

→ Row totals: row1=45, row2=77. Column totals: col1=56, col2=66. Grand total: N=122.

→ Expected counts: $E_{11}=20.66$, $E_{12}=24.34$, $E_{21}=35.34$, $E_{22}=41.66$.

→ $\chi^2 = (17-20.66)^2/20.66 + (28-24.34)^2/24.34 + (39-35.34)^2/35.34 + (38-41.66)^2/41.66 = 1.899$.

Answer: $\chi^2 = 1.899$



Matched-Pairs t-Test

9. A matched-pairs study yields differences: [3, 5, 2, 4, 1, 3, 2, 4]. Test $H_0: \mu_d = 0$ at $\alpha = 0.05$. Compute the t-statistic.

$$d = [3, 5, 2, 4, 1, 3, 2, 4], \quad \bar{d} = 3.0$$

→ Mean of differences: $d\text{-bar} = 3.0$.

→ Standard deviation of differences: $sd = 1.3093$.

→ Standard error: $SE = sd / \sqrt{n} = 1.3093 / \sqrt{7+1} = 0.4629$.

→ $t = d\text{-bar} / SE = 3.0 / 0.4629 = 6.48$.

→ Decision ($df=7$): $|t|=6.48 > 2.093$, reject H_0 at $\alpha=0.05$.

Answer: $t = 6.48, \quad df = 7; \quad |t| = 6.48 > 2.093, \text{reject } H_0 \text{ at } \alpha = 0.05$

10. A matched-pairs study yields differences: [5, 3, -1, 4, 2, 6, 1, 4]. Test $H_0: \mu_d = 0$ at $\alpha = 0.05$. Compute the t-statistic.

$$d = [5, 3, -1, 4, 2, 6, 1, 4], \quad \bar{d} = 3.0$$

→ Mean of differences: $d\text{-bar} = 3.0$.

→ Standard deviation of differences: $sd = 2.2678$.

→ Standard error: $SE = sd / \sqrt{n} = 2.2678 / \sqrt{7+1} = 0.8018$.

→ $t = d\text{-bar} / SE = 3.0 / 0.8018 = 3.74$.

→ Decision ($df=7$): $|t|=3.74 > 2.093$, reject H_0 at $\alpha=0.05$.

Answer: $t = 3.74, \quad df = 7; \quad |t| = 3.74 > 2.093, \text{reject } H_0 \text{ at } \alpha = 0.05$

11. A matched-pairs study yields differences: [5, 3, -1, 4, 2, 6, 1, 4]. Test $H_0: \mu_d = 0$ at $\alpha = 0.05$. Compute the t-statistic.

$$d = [5, 3, -1, 4, 2, 6, 1, 4], \quad \bar{d} = 3.0$$

→ Mean of differences: $d\text{-bar} = 3.0$.

→ Standard deviation of differences: $sd = 2.2678$.

→ Standard error: $SE = sd / \sqrt{n} = 2.2678 / \sqrt{7+1} = 0.8018$.

→ $t = d\text{-bar} / SE = 3.0 / 0.8018 = 3.74$.

→ Decision ($df=7$): $|t|=3.74 > 2.093$, reject H_0 at $\alpha=0.05$.

Answer: $t = 3.74, \quad df = 7; \quad |t| = 3.74 > 2.093, \text{reject } H_0 \text{ at } \alpha = 0.05$

12. A matched-pairs study yields differences: [-2, 1, -3, 0, 2, -1, 1, -2]. Test $H_0: \mu_d = 0$ at $\alpha = 0.05$. Compute the t-statistic.

$$d = [-2, 1, -3, 0, 2, -1, 1, -2], \quad \bar{d} = -0.5$$

→ Mean of differences: $d\text{-bar} = -0.5$.

→ Standard deviation of differences: $sd = 1.7728$.

→ Standard error: $SE = sd / \sqrt{n} = 1.7728 / \sqrt{7+1} = 0.6268$.

→ $t = d\text{-bar} / SE = -0.5 / 0.6268 = -0.8$.

→ Decision ($df=7$): $|t|=0.8 \leq 2.093$, fail to reject H_0 at $\alpha=0.05$.

Answer: $t = -0.8, \quad df = 7; \quad |t| = 0.8 \leq 2.093, \text{fail to reject } H_0 \text{ at } \alpha = 0.05$



Type I and Type II Errors

13. Identify the type of error: H_0 : A new drug has no effect. A researcher rejects H_0 when in fact the drug truly has no effect.

→ H_0 is true but was rejected — this is a Type I error (false positive).

→ Answer: Type I Error.

Answer: *Type I Error*

14. Identify the type of error: H_0 : A machine is working correctly. An inspector fails to reject H_0 even though the machine is actually broken.

→ H_0 is false but was not rejected — this is a Type II error (false negative).

→ Answer: Type II Error.

Answer: *Type II Error*

15. Identify the type of error: H_0 : A student is not cheating. A professor rejects H_0 and accuses the student, but the student was actually innocent.

→ H_0 is true (student is innocent) but was rejected — this is a Type I error (false positive).

→ Answer: Type I Error.

Answer: *Type I Error*

16. Identify the type of error: H_0 : A new drug has no effect. A researcher rejects H_0 when in fact the drug truly has no effect.

→ H_0 is true but was rejected — this is a Type I error (false positive).

→ Answer: Type I Error.

Answer: *Type I Error*

17. Identify the type of error: H_0 : A machine is working correctly. An inspector fails to reject H_0 even though the machine is actually broken.

→ H_0 is false but was not rejected — this is a Type II error (false negative).

→ Answer: Type II Error.

Answer: *Type II Error*

18. Identify the type of error: H_0 : A student is not cheating. A professor rejects H_0 and accuses the student, but the student was actually innocent.

→ H_0 is true (student is innocent) but was rejected — this is a Type I error (false positive).

→ Answer: Type I Error.

Answer: *Type I Error*

19. Identify the type of error: H_0 : A new drug has no effect. A researcher rejects H_0 when in fact the drug truly has no effect.

→ H_0 is true but was rejected — this is a Type I error (false positive).

→ Answer: Type I Error.

Answer: *Type I Error*



20. Identify the type of error: H_0 : A machine is working correctly. An inspector fails to reject H_0 even though the machine is actually broken.

→ H_0 is false but was not rejected — this is a Type II error (false negative).

→ Answer: Type II Error.

Answer: *Type II Error*

21. Identify the type of error: H_0 : A student is not cheating. A professor rejects H_0 and accuses the student, but the student was actually innocent.

→ H_0 is true (student is innocent) but was rejected — this is a Type I error (false positive).

→ Answer: Type I Error.

Answer: *Type I Error*



Two-Sample z-Test for Means

22. Group 1: $\bar{x}_1=95$, $\sigma_1=9$, $n_1=100$. Group 2: $\bar{x}_2=80$, $\sigma_2=14$, $n_2=36$. Test $H_0: \mu_1=\mu_2$ at $\alpha=0.05$.

$$\bar{x}_1 = 95, \bar{x}_2 = 80, \sigma_1 = 9, \sigma_2 = 14, n_1 = 100, n_2 = 36$$

$$\rightarrow SE = \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)} = \sqrt{9^2/100 + 14^2/36} = 2.5009.$$

$$\rightarrow z = (\bar{x}_1 - \bar{x}_2) / SE = (95 - 80) / 2.5009 = 6.0.$$

$$\rightarrow \text{Decision: } |z|=6.0 > 1.96, \text{ reject } H_0 \text{ at } \alpha=0.05.$$

Answer: $z = 6.0$; $|z| = 6.0 > 1.96$, reject H_0 at $\alpha = 0.05$

23. Group 1: $\bar{x}_1=62$, $\sigma_1=5$, $n_1=100$. Group 2: $\bar{x}_2=83$, $\sigma_2=12$, $n_2=49$. Test $H_0: \mu_1=\mu_2$ at $\alpha=0.05$.

$$\bar{x}_1 = 62, \bar{x}_2 = 83, \sigma_1 = 5, \sigma_2 = 12, n_1 = 100, n_2 = 49$$

$$\rightarrow SE = \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)} = \sqrt{5^2/100 + 12^2/49} = 1.7857.$$

$$\rightarrow z = (\bar{x}_1 - \bar{x}_2) / SE = (62 - 83) / 1.7857 = -11.76.$$

$$\rightarrow \text{Decision: } |z|=11.76 > 1.96, \text{ reject } H_0 \text{ at } \alpha=0.05.$$

Answer: $z = -11.76$; $|z| = 11.76 > 1.96$, reject H_0 at $\alpha = 0.05$

24. Group 1: $\bar{x}_1=64$, $\sigma_1=18$, $n_1=64$. Group 2: $\bar{x}_2=79$, $\sigma_2=12$, $n_2=49$. Test $H_0: \mu_1=\mu_2$ at $\alpha=0.05$.

$$\bar{x}_1 = 64, \bar{x}_2 = 79, \sigma_1 = 18, \sigma_2 = 12, n_1 = 64, n_2 = 49$$

$$\rightarrow SE = \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)} = \sqrt{18^2/64 + 12^2/49} = 2.8287.$$

$$\rightarrow z = (\bar{x}_1 - \bar{x}_2) / SE = (64 - 79) / 2.8287 = -5.3.$$

$$\rightarrow \text{Decision: } |z|=5.3 > 1.96, \text{ reject } H_0 \text{ at } \alpha=0.05.$$

Answer: $z = -5.3$; $|z| = 5.3 > 1.96$, reject H_0 at $\alpha = 0.05$



One-sample z-test for a mean

25. Test $H_0: \mu = 48$ against $H_1: \mu \neq 48$. A sample of $n = 16$ gives a sample mean of 44 with $\sigma = 15$. Compute the z test statistic and state the decision at $\alpha = 0.05$.

$$H_0: \mu = 48, \quad \bar{x} = 44, \quad \sigma = 15, \quad n = 16$$

→ Standard error = $\sigma / \sqrt{n} = 15/4 = 3.75$.

→ $z = (\text{sample mean} - \mu_0) / SE = (44 - 48)/3.75 = -1.07$.

→ Compare to ± 1.96 : $|z| = 1.07 < 1.96$, so fail to reject H_0 at $\alpha = 0.05$.

Answer:
$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{44 - 48}{15/4} = -1.07$$

26. Test $H_0: \mu = 114$ against $H_1: \mu \neq 114$. A sample of $n = 25$ gives a sample mean of 105 with $\sigma = 10$. Compute the z test statistic and state the decision at $\alpha = 0.05$.

$$H_0: \mu = 114, \quad \bar{x} = 105, \quad \sigma = 10, \quad n = 25$$

→ Standard error = $\sigma / \sqrt{n} = 10/5 = 2.0$.

→ $z = (\text{sample mean} - \mu_0) / SE = (105 - 114)/2.0 = -4.5$.

→ Compare to ± 1.96 : $|z| = 4.5 > 1.96$, so reject H_0 at $\alpha = 0.05$.

Answer:
$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{105 - 114}{10/5} = -4.5$$

27. Test $H_0: \mu = 49$ against $H_1: \mu \neq 49$. A sample of $n = 100$ gives a sample mean of 37 with $\sigma = 15$. Compute the z test statistic and state the decision at $\alpha = 0.05$.

$$H_0: \mu = 49, \quad \bar{x} = 37, \quad \sigma = 15, \quad n = 100$$

→ Standard error = $\sigma / \sqrt{n} = 15/10 = 1.5$.

→ $z = (\text{sample mean} - \mu_0) / SE = (37 - 49)/1.5 = -8.0$.

→ Compare to ± 1.96 : $|z| = 8.0 > 1.96$, so reject H_0 at $\alpha = 0.05$.

Answer:
$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{37 - 49}{15/10} = -8.0$$



One-Sample z-Test for a Proportion

28. A researcher claims the true proportion is $p_0 = 0.5$. A sample of $n = 236$ gave 157 successes. Test $H_0: p = 0.5$ at $\alpha = 0.05$.

$$H_0 : p = 0.5, \quad \hat{p} = 0.6653, \quad n = 236$$

→ Sample proportion: $p\text{-hat} = 157/236 = 0.6653$.

→ Standard error under H_0 : $SE = \sqrt{0.5 \cdot (1-0.5)/236} = 0.0325$.

→ $z = (p\text{-hat} - p_0) / SE = (0.6653 - 0.5) / 0.0325 = 5.08$.

→ Decision: $|z|=5.08 > 1.96$, reject H_0 at $\alpha=0.05$.

Answer: $z = 5.08$; $|z| = 5.08 > 1.96$, reject H_0 at $\alpha = 0.05$

29. A researcher claims the true proportion is $p_0 = 0.6$. A sample of $n = 143$ gave 87 successes. Test $H_0: p = 0.6$ at $\alpha = 0.05$.

$$H_0 : p = 0.6, \quad \hat{p} = 0.6084, \quad n = 143$$

→ Sample proportion: $p\text{-hat} = 87/143 = 0.6084$.

→ Standard error under H_0 : $SE = \sqrt{0.6 \cdot (1-0.6)/143} = 0.041$.

→ $z = (p\text{-hat} - p_0) / SE = (0.6084 - 0.6) / 0.041 = 0.21$.

→ Decision: $|z|=0.21 \leq 1.96$, fail to reject H_0 at $\alpha=0.05$.

Answer: $z = 0.21$; $|z| = 0.21 \leq 1.96$, fail to reject H_0 at $\alpha = 0.05$

30. A researcher claims the true proportion is $p_0 = 0.7$. A sample of $n = 453$ gave 149 successes. Test $H_0: p = 0.7$ at $\alpha = 0.05$.

$$H_0 : p = 0.7, \quad \hat{p} = 0.3289, \quad n = 453$$

→ Sample proportion: $p\text{-hat} = 149/453 = 0.3289$.

→ Standard error under H_0 : $SE = \sqrt{0.7 \cdot (1-0.7)/453} = 0.0215$.

→ $z = (p\text{-hat} - p_0) / SE = (0.3289 - 0.7) / 0.0215 = -17.24$.

→ Decision: $|z|=17.24 > 1.96$, reject H_0 at $\alpha=0.05$.

Answer: $z = -17.24$; $|z| = 17.24 > 1.96$, reject H_0 at $\alpha = 0.05$

